

REPOSITORY 04 · EXTENDED DECK

AI, Human Cognition and Knowledge Collapse

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[repository link](#)

Roadmap: result first, then the stress test

1 · Static core

Signals, effort, Observation 1, best response.

2 · Dynamics

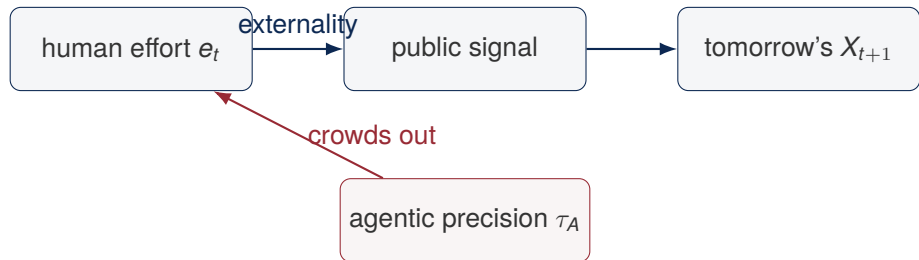
Propositions 1–2 and the route to collapse.

3 · Audit

$\Delta_I > 0$, endpoints, welfare, Lean.

The persuasive result is the sign logic.
The fragile result is the literal zero endpoint.

The cognitive commons

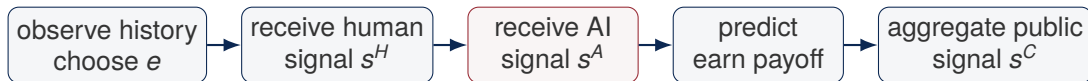


Verdict

The private agent does not internalize the arrow from effort to tomorrow's public stock.

Source: Introduction and Sections 3.1–3.3.

Within-period timing



Commitment margin

Effort is chosen before either private signal arrives. The current agent cannot change inherited X_t .

Source: Section 3.1, timing (i)–(v).

Two states, three signals

Object	Target	Precision	Who benefits?
Prior/context	$\theta_{i,t}$	σ^{-2}	agent i
Human private $s_{i,t}^H$	$\theta_{i,t}$	$\lambda_I \mathbf{e}_{i,t}$	agent i
Agentic $s_{i,t}^A$	$\theta_{i,t}$	τ_A	agent i
Community s_t^C	common θ_t	$\lambda_G E_t$	future cohorts

$$Y_{i,t} = \sigma^{-2} + \lambda_I \mathbf{e}_{i,t} + \tau_A, \quad X_t = \text{Var}(\theta_t \mid \mathcal{I}_t)^{-1}.$$

Source: Equations (1)–(4).

Production turns precision into success

$$f(\mathbf{1}\{|x - \theta| \leq 1\}, \mathbf{1}\{|y - \theta| \leq 1\}), \quad f : \{0, 1\}^2 \rightarrow \mathbb{R}.$$

Payoff increments

$$\begin{aligned}\Delta_G &= f_{10} - f_{00}, & \Delta_I &= f_{01} - f_{00}, \\ \Delta_X &= f_{11} - f_{10} - f_{01} + f_{00}.\end{aligned}$$

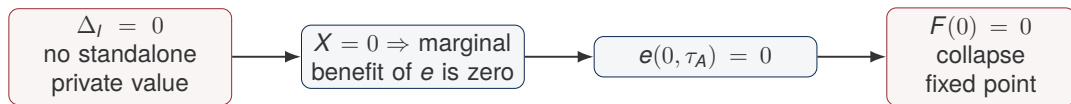
Baseline production hinge

$$\Delta_I = 0, \quad \Delta_X > 0.$$

Context-specific correctness has no value unless the common state is also predicted correctly.

Source: Section 3.1, Assumption 1.

What Assumption 1 buys



This chain is logical—but it is not robust to $\Delta_I > 0$.

The agent's optimization problem

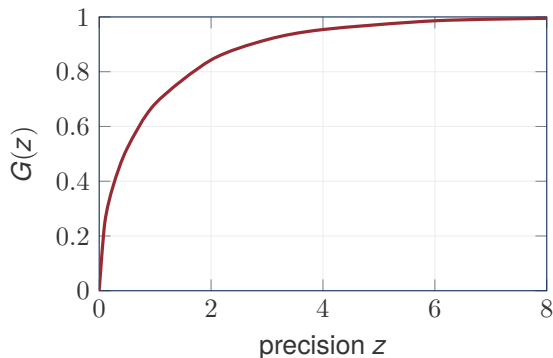
$$\max_{e \geq 0} U(e; X, \tau_A) = f_{00} + G(X)\Delta_G + G(X)G(Y)\Delta_X - \frac{\varepsilon}{\varepsilon + 1} e^{(\varepsilon+1)/\varepsilon},$$
$$Y = \sigma^{-2} + \lambda_I e + \tau_A, \quad G(z) = 2\Phi(\sqrt{z}) - 1.$$

What is taken as given?

X is inherited; τ_A is technology; the atomistic agent ignores her infinitesimal effect on public learning.

Source: Equation (6) and Section 3.5.

From Gaussian precision to a payoff



$$G(z) = \Pr(|Z| \leq 1), \quad Z \sim N(0, z^{-1}).$$

- ▶ $G'(z) = g(z) > 0$;
- ▶ $g'(z) < 0$;
- ▶ extra precision helps, with diminishing returns.

Boundary

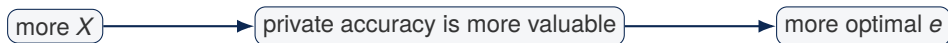
G is continuous at zero, but its right slope is unbounded.

Source: Section 3.3.

General knowledge complements effort

$$U_e = \Delta_X G(X) \lambda_I g(Y) - e^{1/\varepsilon}.$$

$$U_{eX} = \Delta_X \lambda_I g(X) g(Y) > 0 \quad \text{for } X > 0.$$



Source: Observation 1, first mixed partial.

Agentic precision substitutes for effort

$$U_{e\tau_A} = \Delta_X G(X) \lambda_I g'(Y) < 0.$$



Verdict

This is a precision-substitution result. The mathematics does not require autonomy or agency.

Source: Observation 1, second mixed partial.

Observation 1: exact conditions

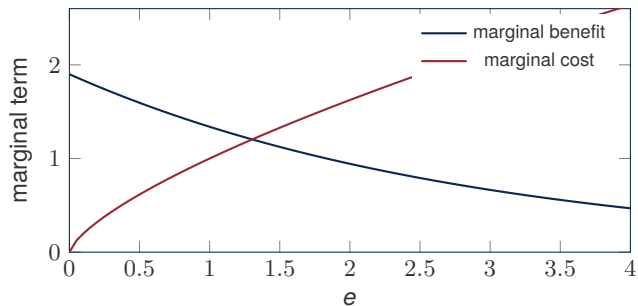
Condition	Role
$X > 0, Y > 0$	ordinary interior derivatives $g(X)$ and $g'(Y)$
$\lambda_I > 0$	effort truly raises private precision
$\Delta_X > 0$	success in the two dimensions is complementary
$\Delta_I = 0$	baseline payoff formula and zero-effort boundary
Gaussian independent errors	converts correct-prediction events to $G(X)G(Y)$
$\tau_A \geq 0, \mathbf{e} \geq 0, \varepsilon > 0$	feasible technology and convex cost

One-sentence intuition

Shared knowledge makes private learning useful; AI already supplies the same private precision that effort would buy.

A unique best response

$$\Delta_X G(X) \lambda_I g(\sigma^{-2} + \lambda_I e + \tau_A) = e^{1/\varepsilon}.$$



Source: Section 3.5; schematic, not a calibration.

Observation 2: optimizer comparative statics

Raise inherited X

$U_{eX} > 0$ shifts marginal benefit upward, so $e(X, \tau_A)$ rises.

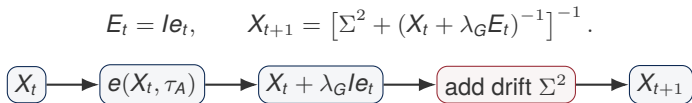
Raise agentic τ_A

$U_{e\tau_A} < 0$ shifts marginal benefit downward, so $e(X, \tau_A)$ falls.

$$\begin{aligned} X_2 > X_1 > 0 &\Rightarrow e(X_2, \tau_A) > e(X_1, \tau_A), \\ \tau'_A > \tau_A &\Rightarrow e(X, \tau'_A) < e(X, \tau_A) \quad (X > 0). \end{aligned}$$

Source: Observation 2, Topkis/strict single-crossing logic.

The Kalman law of motion



Source: Equations (2), (3), and symmetric equilibrium.

Proposition 1: dynamic equilibrium

Exact selected statement

Fix $X_1 \geq 0$ implied by the prior. There exists a unique symmetric Perfect Bayesian equilibrium path with $e_{i,t} = e_t \geq 0$, where

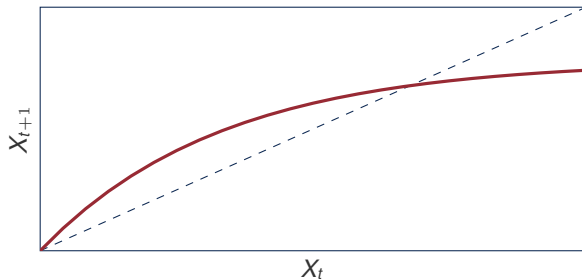
$$e_t = e(X_t, \tau_A), \quad X_{t+1} = [\Sigma^2 + (X_t + \lambda_G I e_t)^{-1}]^{-1}.$$

Conditions doing the work: atomistic agents, equal-sized islands, Gaussian signals, unique static best response, common parameters, and a fixed AI precision.

Source: Proposition 1, equation (7).

The one-dimensional transition map

$$F(X; l, \tau_A) = [\Sigma^2 + \{X + \lambda_G l e(X, \tau_A)\}^{-1}]^{-1}.$$



Source: Equation (8); schematic only.

Proposition 2: comparative statics of F

Statement

F is pointwise increasing in aggregation scale I and pointwise decreasing in agentic precision τ_A , strictly for all $X > 0$.

$$I \uparrow \Rightarrow Ie(X, \tau_A) \uparrow \Rightarrow F(X) \uparrow .$$

$$\tau_A \uparrow \Rightarrow e(X, \tau_A) \downarrow \Rightarrow F(X) \downarrow .$$

Verdict

This proposition orders tomorrow's public precision. It does not alone establish welfare loss or global collapse.

Source: Proposition 2, p. 17.

Why the number four appears

Near $X = 0$, the model compares

$$e(X, \tau_A) = \Theta(X^{\varepsilon/2}) \quad \text{with} \quad e_{\text{maint}}(X) = \Theta(X^2).$$

$$\varepsilon < 4$$

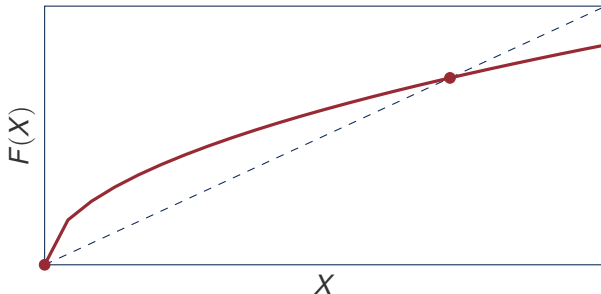
$\varepsilon/2 < 2$: chosen effort vanishes more slowly; zero is locally unstable.

$$\varepsilon > 4$$

$\varepsilon/2 > 2$: chosen effort vanishes faster; zero is locally stable.

Source: Lemma 2 and discussion, pp. 17–18.

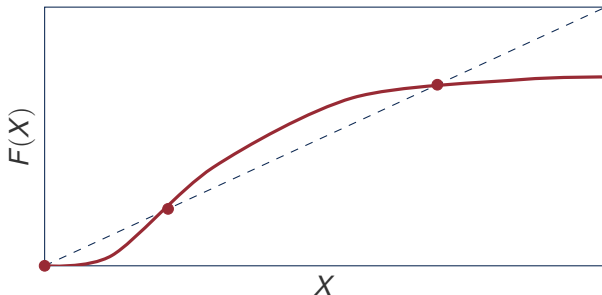
Robust-learning regime: $\varepsilon < 4$



Proposition 3: exactly two steady states, $0 = \bar{X}_\ell < \bar{X}_h$; zero is unstable, $\bar{X}_h$ stable, and every $X_1 > 0$ converges to $\bar{X}_h$.

Source: Proposition 3; schematic only.

Collapse regime: $\varepsilon > 4$



If $\tau_A < \tau_A^c$, low and high steady states coexist around an unstable threshold; if $\tau_A > \tau_A^c$, only zero remains. The proposition does not state the equality case.

Source: Proposition 5 and Figure 2; schematic only.

Is welfare increasing in AI accuracy?

$$\frac{\partial \bar{U}^+}{\partial \tau_A} = \underbrace{g(\bar{Y}_h)G(\bar{X}_h)\Delta_X}_{\text{direct gain} \geq 0} + \underbrace{\frac{\partial G(\bar{X}_h)}{\partial \tau_A} [\Delta_G + G(\bar{Y}_h)\Delta_X]}_{\text{dynamic loss} \leq 0}.$$

Answer

Not unconditionally. Single-peakedness needs $\sigma^{-2} \geq \sqrt{2} - 1$ (Assumption 2) and the appropriate high-knowledge steady state; with $\varepsilon > 4$, basin selection is an additional channel.

Source: Section 4.3, Assumption 2, Propositions 10–11.

What the authors relax in Section 5

Extension	Change	What survives
AI aggregation Synthetic data Effort separability	$I(\tau_A) = I_0 + e^{\eta\tau_A}$ add public precision τ_{syn} public input e^β	collapse force if $\eta < \varepsilon/2$ zero becomes a positive floor cutoff changes to $\varepsilon < 4/\beta$
Never relaxed	$\Delta_I = 0$	literal zero-collapse hinge

Source: Sections 5.1–5.3, Propositions 14–16.

New extension: give private knowledge standalone value

$$\Delta_I > 0 \Rightarrow U_e = \lambda_I g(Y) [\Delta_I + G(X) \Delta_X] - e^{1/\varepsilon}.$$



Verdict

A low-knowledge trap may remain, but exact zero is not a fixed point. This is a production-side robustness failure, not a numerical curiosity.

Lean: what is actually verified

Mathematical claim

$$\Delta_I > 0, \lambda_I > 0, g(Y) > 0 \Rightarrow \lambda_I g(Y) \Delta_I > 0.$$

At $e = X = 0$, the marginal benefit is positive; a zero-effort optimizer is impossible under the paper's monotone marginal structure.

Lean boundary

```
theorem positiveStandaloneReturn
  (hDI : 0 < deltaI)
  (hLI : 0 < lambdaI)
  (hg : 0 < gY) :
  0 < lambdaI * gY * deltaI := by
  positivity
```

Interpretation

Lean verifies the ordered-ring implication. EconCSLib separately audits whether this statement faithfully represents the paper and extension.

Source: Repository 'lean/'; final theorem name may follow the generated interface.

Takeaways: what survives the audit

I believe

- ▶ the static mixed-partial signs on $X > 0$;
- ▶ the pointwise transition-map comparisons;
- ▶ the externality as a coherent cautionary mechanism.

I qualify

- ▶ “agentic” is interpretation, not needed for the sign;
- ▶ literal collapse relies on $\Delta_I = 0$;
- ▶ welfare needs Assumption 2 and equilibrium selection;
- ▶ $X = 0$ and $\tau_A = \tau_A^C$ require explicit endpoint treatment.

The robust message is a **low-knowledge trap**,
not an unconditional theorem that better AI destroys knowledge.